

Question	Marking Guidance	Mark	Comments
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02.1

All four rows correct = **2 marks**;;

3 or 2 rows correct = **1 mark**;

Statement	Part of the nephron			
	Glomerulus	Proximal convoluted tubule	Distal convoluted tubule	Collecting duct
ADH receptors are present			✓	✓
Water can be absorbed from the filtrate		✓	✓	✓
Blood is under high pressure	✓			
Glucose is absorbed from the filtrate		✓		

2
(2 x
AO1)

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02.2	1. (Plasma) proteins are (too) large;	2 (2 x AO1)	2. Ignore any other structures
	2. To filter/pass through <u>basement membrane</u> ;		

Question	Marking Guidance	Mark	Comments
02.3	Max 4 for MP3-7 (Yes – no mark) 1. Probability of less than 0.01 that the increase (in GS) is due to chance OR Increase (in GS) is significant; 2. Rats and humans are both mammals, so similar; (No – no mark) 3. (Investigation only) done on rats OR (Investigation) not done on humans; 4. Do not know if (160-day-old) rats are adults; 5. Small sample size, so not representative; 6. Only two (different) quantities of protein (used) OR Only high and normal protein (milk/s) OR Protein content (of milk/s) not known OR Types of protein (in the milk/s) not known; 7. Only GS studied OR Other kidney damage (can) occur/s;	5 max (5 x AO3)	1. Accept probability of less than 1%/0.05/5% that the increase is due to chance 1. Accept probability of more than 99%/0.99/95%/0.95 that the increase is not due to chance 1. Accept P for probability 1. Accept difference for increase 1. Ignore correlation is significant 1. Ignore results are significant unqualified 4. Ignore long terms effects unknown 5. Accept only 14/28 (rats), so not representative 6. Ignore answers relating to protein content of solid food diet 7. Accept other named factors (can cause kidney damage) e.g. genetic factors
